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Analysis of Coupling Mechanism and Research on Control Strategy for Rotorcraft Aerial Manipulators

LingJu Kong, Pengjun Mao*, Chengyuan Zhan, Ashutosh Sharma, and Kai Guo*

Abstract—Recently, automation using Aerial Manipulators (AM) plays a vital role in the consumer centric industries where tasks need to be accurate with stability. Rotorcraft aerial manipulators are hybrid robotic systems composed of multi-rotor unmanned aerial vehicles (UAVs) and robotic arms. They possess advantages such as high-altitude flexible maneuverability and interactive operation, rendering significant application value in fields like high-altitude maintenance, hazardous material transportation, and emergency rescue. However, the strong nonlinear force-position coupling between the UAV and the robotic arm during operation leads to unstable flight attitudes and increased positioning errors at the end-effector of the robotic arm. Traditional PID control methods struggle to cope with such complex coupling characteristics. To address this critical issue, this paper first reveals the chain coupling mechanism of "centroid offset-inertial transfer-aerodynamic disturbance" for rotorcraft aerial manipulators from three dimensions combined with dynamic simulations. On this basis, a multi-field coupling dynamic equation covering centroid offset, inertial coupling, and aerodynamic disturbance is innovatively established, providing a theoretical foundation for the design of control strategies. Furthermore, an adaptive PID control strategy based on multi-agent reinforcement learning (MAAC-PID) is proposed, which innovatively adopts a "centralized learning-distributed execution" architecture. Through two agents optimizing the UAV's PID parameters and the robotic arm's joint control respectively, global cooperative optimization of the coupled system is achieved. Simulation results show the MAAC-PID algorithm outperforms traditional PID and PPO, reducing trajectory errors and enhancing stability. Specifically, under 2KG load condition, the standard deviation of the error of the MAAC-PID algorithm in the square trajectory is reduced by approximately 16.1% compared with the PPO algorithm; in addition, the MAAC-PID algorithm achieves an improvement of 0.03m in average maximum trajectory error compared with the PPO algorithm, exhibiting stronger robustness and adaptability.

Index Terms—Rotorcraft Aerial Manipulator, Force-Position Coupling, Multi-Agent Reinforcement Learning, Control Strategy

I. INTRODUCTION

In recent years, the e-commerce and consumer electronics industries have seen an exponential growth in terms of services and efficiency. There are several important factors which are strengthening this trend. One of the important factors is heavy use of UAV and robotics systems for automation and

services [1]. The UAVs are contributing at different levels as per the task requirements in different areas such as smart e-business, smart agriculture, smart transportation, and others [2-4]. In addition to this, usage of robotics in the industries also plays an important role along with UAV as an actor element. With passage of time, while understanding the amalgamation of UAV and robotics in different applications, advancement in the UAV and robotic technology has been experienced. UAV technology has been advanced along with robotic system known as AM or Aerial Robotic Manipulator. As a hybrid robotic system, rotorcraft aerial manipulators integrate the high-altitude maneuverability of multi-rotor UAVs with the dexterous and interactive operational capabilities of robotic arms. They provide innovative solutions for tasks in high-altitude, high-risk, and complex environments. Demonstrating unparalleled flexibility and safety compared to traditional operation methods, this system has become one of the most focused research hotspots in the field of robotics and automation over the past decade [5]. Relevant technological breakthroughs hold significant practical significance for industrial production and emergency support.

Currently, research on rotorcraft aerial manipulators mainly focuses on three core directions: control algorithm design [6-8], dynamic modeling and analysis [9-10], and perception and interaction technology development [11-16]. Among these, high-performance control algorithms are the key to achieving precise operations. In the research of control algorithms, Liang et al. proposed a physical interaction control method based on external torque estimation, which effectively improved the stability and operational accuracy of rotorcraft aerial manipulators during environmental interaction [6]; Chen et al. designed a nonsingular terminal sliding mode adaptive control strategy, successfully realizing dynamic grasping tasks of rotorcraft aerial manipulators [7]; Lunni et al. adopted nonlinear model predictive control (NMPC) to achieve efficient operation scheduling of rotorcraft aerial manipulators [8]. In the field of dynamic modeling and analysis, Xu et al. combined visual feedback with dynamic models to achieve precise control of contact force by real-time adjusting impedance control parameters [9]; Hu et al. proposed a hierarchical impedance control method, realizing precise control of rotorcraft aerial manipulators in different operational phases through a layered control strategy [10]. In

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terms of perception and interaction technology, Kim et al. constructed a human-machine interaction channel through haptic feedback technology, enabling operators to precisely remotely control rotorcraft aerial manipulators [13]; ALLENSPACH et al. proposed a 6-degree-of-freedom bilateral teleoperation method, expanding the application scenarios of rotorcraft aerial manipulators in physical interaction tasks [15].

Despite the certain progress made in existing research, significant limitations exist, making it difficult to meet the high-precision control requirements under complex working conditions. Specifically, the inertial force or torque generated by the movement of the robotic arm will act back on the UAV; while when the UAV accelerates, each link of the robotic arm will be subjected to inertial force, which is transmitted to the end-effector through the joints [17]. This coupling effect will lead to unstable flight attitudes of the rotorcraft aerial manipulator and increased end-effector positioning errors, and may even cause UAV rollover or drift in extreme cases, seriously restricting the operational efficiency of the rotorcraft aerial manipulator. Therefore, developing a cooperative control strategy with dynamic decoupling capability has become a key problem urgently to be solved in the current field.

In recent years, Multi-Agent Reinforcement Learning (MRL) has demonstrated excellent performance in dynamic game and distributed decision-making fields [18-20]. Through the competition and cooperation mechanisms among agents to achieve strategy cooperative optimization, it provides a new idea for solving the complex coupling control problem of rotorcraft aerial manipulators [21]. However, most existing MRL-related research focuses on single robotic arm or fixed base scenarios, and the MRL cooperative control strategies for rotorcraft aerial manipulators still have significant limitations: insufficient model generalization ability to adapt to the dynamically changing operational conditions of rotorcraft aerial manipulators; low decision-making efficiency in communication-constrained scenarios, failing to meet real-time control requirements. Meanwhile, traditional control methods generally have problems such as incomplete decoupling and insufficient robustness when dealing with the strong nonlinearity and strong coupling characteristics of force-position coupling in rotorcraft aerial manipulators. Therefore, developing a rotorcraft aerial manipulator controller with both precision and robustness remains a research hotspot and difficulty in the current field.

To address the above issues, this paper proposes an adaptive PID control strategy based on multi-agent reinforcement learning (MAAC-PID) on the basis of systematically analyzing the coupling mechanism of rotorcraft aerial manipulators. The research content mainly includes three parts: first, combining core characteristics such as centroid offset and inertial coupling, construct a dynamic model of the rotorcraft aerial manipulator, and quantitatively analyze the influence laws of coupling forces and torques on the flight attitude and stability of the rotorcraft aerial manipulator through dynamic simulation experiments; second, design the MAAC-PID control strategy, organically integrating MRL with adaptive PID control to achieve system decoupling and stable control; finally, verify the effectiveness

of the proposed control strategy through simulation experiments.

The main contributions of this paper are as follows:

- i. For the first time, a multi-field coupling dynamic model of rotorcraft aerial manipulators is established. Different from existing studies focusing on local coupling effects, this paper reveals the chain coupling law of "centroid offset-inertial transfer-aerodynamic disturbance" from three dimensions (mechanical structure configuration, flight attitude adjustment, and robotic arm movement) combined with dynamic simulations, and derives a dynamic equation that can fully reflect multi-field coupling effects, providing a solid theoretical support for the design of control strategies.
- ii. Innovatively propose the MAAC-PID strategy with dual-agent cooperative optimization: innovatively integrating multi-agent reinforcement learning with adaptive PID control, through the dynamic cooperation between the UAV attitude control agent and the robotic arm joint control agent, breaking through the problems such as "incomplete decoupling" of traditional control and "insufficient generalization" of single-agent reinforcement learning. Compared with traditional PID control and PPO algorithms, the average trajectory error is significantly reduced.
- iii. Expand the application boundary of MRL in dynamically coupled systems: through the "centralized learning-distributed execution" architecture, expand the applicable scenario of MRL to the dynamically coupled scenario of rotorcraft aerial manipulators. Simulation verification shows that the MAAC-PID control strategy can meet the real-time operational requirements, providing a new path for the engineering application of such hybrid robots.

The structure of this paper is as follows: Section 1 introduces the research status of aerial manipulators and the main research content of this paper; Section 2 presents the dynamic model of aerial manipulators and analysis of the force-position coupling mechanism; Section 3 elaborates on the MAAC-PID control strategy based on multi-agent reinforcement learning and the decoupling method; Section 4 verifies the effectiveness of the proposed control strategy and decoupling method through simulation experiments; Section 5 summarizes the research conclusions and prospects future research directions.

II. ANALYSIS OF COUPLING MECHANISM IN AERIAL MANIPULATOR SYSTEMS

A rotorcraft aerial manipulator is a complex system composed of a rotor UAV and a robotic arm. This system involves multiple physical interactions among the UAV, robotic arm, end-effector, and its surrounding environment, exhibiting dynamic characteristics such as multi-input multi-output (MIMO), high nonlinearity, and strong coupling [22]. The system's structural features, attitude adjustments during flight, and the movement of the robotic arm itself collectively lead to continuous changes in the system's centroid position, inertial parameters, aerodynamic characteristics, and control allocation matrix, thereby triggering complex multi-physics coupling effects. This effect renders the system dynamics significantly nonlinear, directly restricting its operational accuracy and overall stability

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[23]. In this section, we take the coupling impacts of three factors—system structural features, flight attitude, and robotic arm movement—as the entry point. Combined with simulation experiments, we conduct an in-depth analysis of the coupling mechanism and establish a system coupling dynamic model suitable for this research.

A. Analysis of System Mechanical Structure Coupling

1) Mechanism and Modeling of Mechanical Structure Coupling

The coupling effect induced by the mechanical structure mainly stems from the centroid offset caused by system assembly and specific configuration. Even when the system is in an initial static or hovering state, this offset introduces inherent static coupling, which serves as one of the fundamental and primary causes of system coupling. In this state, although the force and torque exerted on the robotic arm base remain constant and the system's flight attitude stays stable, the centroids of the UAV and the robotic arm cannot coincide and are usually not strictly aligned along the z-axis of the rotorcraft aerial manipulator coordinate system. Consequently, a continuous centroid offset and corresponding static coupling forces and torques are generated.

To focus on the core mechanism of structural coupling and simplify the analysis, the following modeling assumptions are established:

- i. The UAV is regarded as an ideal rigid body: The UAV (including the frame) is treated as a symmetric rigid body with uniform mass distribution, where the geometric center coincides with the centroid.
- ii. Constraints on connection and motion: The robotic arm forms a rigid connection with the UAV through the base, and its working space is limited to the robotic arm's operational domain below the base.
- iii. Simplification of the external environment: The resistance and gravity acting on the UAV during flight remain constant.
- iv. Each link of the robotic arm is a rigid structure without elastic deformation;
- v. Initial configuration of the robotic arm: The initial angles of the robotic arm joints are as follows: joint1 -70.0°, joint2 61.0°, joint3 12.0°.

Based on the above assumptions, mathematical modeling is performed. Let m_a be the mass of the UAV with centroid O_a , and m_b be the mass of the robotic arm with centroid O_b . The total mass of the system is $m = m_a + m_b$. In the body coordinate system, the position vector of the centroid of the i -th link of the robotic arm relative to O_a is $r_i = [x_i, y_i, z_i]^T$. Then, the position vector of the overall centroid O_b of the robotic arm relative to

$$O_a \text{ is } r_b = \frac{1}{m_b} \sum_{i=1}^n m_{bi} r_i.$$

Based on the principle of mass conservation:

$$m \cdot r_c = m_a \cdot 0 + m_b \cdot r_b$$

Then, the centroid offset of the system can be expressed as:

$$r_c = [x_c, y_c, z_c]^T = \frac{1}{m} \sum_{i=1}^n m_{bi} r_i$$

The centroid offset caused by the mechanical structure and the initial state configuration of the robotic arm induces coupling forces during flight:

$$\Delta F = \sum_{k=1}^4 (F_k - \frac{mg}{4}) u_z \quad (1)$$

Wherein, F_k denotes the total rotor lift, and $u_z = [0, 0, 1]^T$, represents the unit vector along the positive z-axis. When the system has a linear acceleration a , two coupling torques are generated: the coupling torque M_G caused by centroid offset and the coupling torque M_I induced by inertial force:

$$\Delta M = M_G + M_I = mg \cdot r_c + (-ma) \cdot r_c$$

$$= m \begin{vmatrix} i & j & k \\ x_c & y_c & z_c \\ 0 & 0 & mg \end{vmatrix} - m \begin{vmatrix} i & j & k \\ x_c & y_c & z_c \\ ax & 0 & 0 \end{vmatrix} \quad (2)$$

The inherent centroid offset caused by the mechanical structure is the physical root cause of the fundamental coupling effects in the system under both static and dynamic conditions. Equations (1) to (2) quantitatively describe the static and inertial coupling effects induced by the structure, providing a key model basis for subsequent dynamic analysis and the design of decoupling control strategies.

2) Coupled Dynamic Simulation Analysis of Mechanical Structures

To visually verify the aforementioned theoretical analysis and quantify the coupling effect, this study constructs a high-fidelity simulation model of the rotor flying manipulator as illustrated in Fig. 1, based on the Gazebo simulation environment. The system consists of a quadrotor unmanned aerial vehicle (UAV) and a 3-degree-of-freedom (3-DOF) manipulator connected via a rigid joint. The flight controller employs a standard PID control architecture based on the PX4 firmware, and is equipped with a Global Positioning System (GPS), Inertial Measurement Unit (IMU), magnetometer, barometer, and six-axis torque sensor to simulate real-world sensory feedback. The key parameters of the UAV and the manipulator are listed in the following text and Table I, respectively.

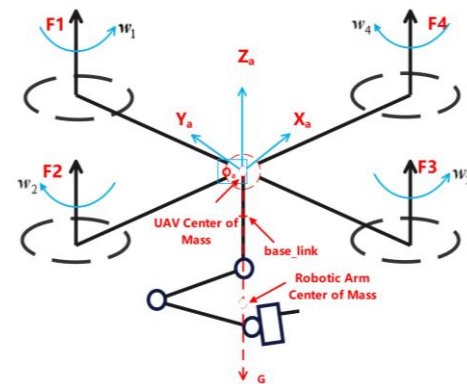


Fig. 1. Schematic diagram of the rotor flying manipulator structure.

Parameters of the Quadrotor Unmanned Aerial Vehicle (UAV): Rotor radius: 0.128 m, Total fuselage mass: 1.5 kg, Fuselage dimensions: 0.47 × 0.47 × 0.47 m.

Parameters of the manipulator are listed in Table I.

Table I: Parameters of the manipulator

Name	Length (m)	Weight (kg)	Initial Angle
Base link (Base)	0.05	0.3	0
Joint 1	0.12	0.25	-70.0°
Link 1(Shoulder)	0.13	0.15	-

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Joint 2	0.12	0.2	61.0°
Link 2(Elbow)	0.13	0.12	-
Joint 3	0.05	0.1	12.0°
Gripper (Wrist)	0.06	0.05	-

The simulation experiment is conducted with the system operating in an initial hovering state, aiming to monitor the flight attitude angles (roll, pitch, yaw) as well as the forces and torques acting on the manipulator base. This serves to verify the rationality of the structural coupling analysis.

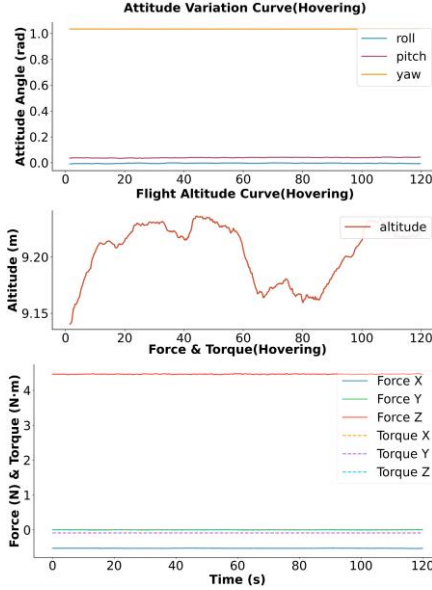


Fig. 2. Flight attitude and forces/torques acting on the base under hovering state based on PID.

The simulation results are presented in Fig. 2. Under the hovering state, both the roll and pitch angles stabilize at approximately 0°, while the yaw angle maintains the initially set value of 1.03 rad, indicating the overall attitude stability of the system. However, data analysis of the six-axis force/torque sensor mounted at the manipulator base reveals that, in addition to the lift force counteracting gravity along the Z-axis, a lateral force of approximately 0.53 N persists along the X-axis, and slightly fluctuating forces and torques are also observed along the Y-axis. This phenomenon is in full agreement with the predictions of theoretical analysis, directly validating the coupled correlation mechanism of "mechanical structure configuration → intrinsic center of mass (CoM) offset → steady-state disturbance forces/torques." The quantitative results provide direct basis and optimization objectives for the subsequent design of targeted adaptive compensation control strategies, which is crucial for improving the attitude control accuracy and overall reliability of the system during precise operations.

B. Coupling Mechanism Induced by Flight Attitude Changes

The meanings of the symbols in the formula are listed in Table II.

Table II: Nomenclature

Symbol	Description
F_u	Total coupling force acting on the UAV fuselage
n	Number of manipulator links
m_i	Mass of the i-th manipulator link

$\dot{V}_u$	Linear acceleration of the UAV's center of mass (CoM) in the body coordinate system relative to the inertial coordinate system
ω	Angular velocity vector of the UAV's body coordinate system relative to the inertial coordinate system
$\dot{\omega}$	Angular acceleration vector of the UAV's body coordinate system relative to the inertial coordinate system
r_i	Position vector of the CoM of the i-th manipulator link relative to the UAV's CoM
$\theta, \dot{\theta}, \ddot{\theta}$	Joint angles, angular velocities, and angular accelerations of the manipulator
$J_{r,i}(\theta)$	Linear velocity Jacobian matrix of the i-th link
$\dot{J}_{r,i}(\theta, \dot{\theta})\dot{\theta}$	Time derivative of $J_{r,i}(\theta)$

1) Coupling Analysis and Modeling of Flight Attitude Changes

The dynamic adjustment of flight attitude is a key factor inducing strong coupling. Attitude changes (roll, pitch, yaw) directly result in time-varying offsets of the system's CoM and alter the aerodynamic force distribution. Taking pitch as an example, changes in the fuselage angle not only generate disturbance forces along the body axes but also introduce disturbance torques around the body axes due to the redistribution of rotor lift vectors and the system's own inertia. These disturbance forces/torques are transmitted to the manipulator through the base, interfering with its kinematic and dynamic states, and ultimately manifest as end-effector positioning errors.

$$r_{inst}^{\Gamma}(q, \theta) = \frac{m_{UAV} r_{UAV}^{\Gamma} + m_{arm} r_{arm}^{\Gamma}(q)}{m}$$

Where q denotes the joint angle vector, $m = m_{UAV} + m_{arm}$ (with m_{UAV} and m_{arm} representing the masses of the UAV and the manipulator, respectively), and θ is the UAV attitude angle.

Assuming the angular velocity of the quadrotor UAV in the body coordinate system is $\omega = [\omega_x, \omega_y, \omega_z]^T$, m_{end} denotes the mass of the end-effector, and r_{end}^B represents the coordinate of the manipulator's end-effector in the body coordinate system, the inertial force acting on the manipulator's end-effector can be expressed as:

$$F_{inst}^I = -m_{end} \left[(\dot{\omega} \times r_{end}^B) + \omega \times (\omega \times r_{end}^B) \right] \quad (3)$$

2) Positional Coupling Effects of Flight Attitude Changes on the End-Effector

Flight attitude changes mainly induce positional errors through "attitude deviations" and "inertial forces due to angular acceleration", comprehensively affecting the operational accuracy of the end-effector.

i. Position error induced by attitude deviations

When the system exhibits attitude deviations ($\Delta\phi, \Delta\theta, \Delta\psi$), the actual position of the end effector in the inertial frame can be expressed as:

$$P_E^I = P_B^I + R_B^I(\Delta\phi, \Delta\theta, \Delta\psi) \cdot P_E^B$$

Where P_E^I denotes the position of the UAV's center of mass (CoM) in the inertial coordinate system, P_E^B represents the position of the end-effector relative to the UAV's CoM, and R_B^I is

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the rotation matrix.

The positional error of the end-effector induced by attitude changes can be expressed as:

$$\Delta P_E^I = P_E^I + R_B^I(\Delta\phi, \Delta\theta, \Delta\psi) \cdot P_E^B - (P_E^I + P_E^B)$$

$$= [R_B^I(\Delta\phi, \Delta\theta, \Delta\psi) - I] \cdot P_E^B$$

Where I denotes the identity matrix (I).

When attitude deviations are small, $\cos\alpha \approx 1$ and $\sin\alpha \approx \alpha$ (where α represents the attitude angle deviation), and the rotation matrix R_B^I can be simplified as:

$$R_B^I \approx I + \begin{bmatrix} 0 & -\Delta\psi & \Delta\theta \\ \Delta\psi & 0 & -\Delta\phi \\ -\Delta\theta & \Delta\phi & 0 \end{bmatrix}$$

From the above approximation, the linear expression of the positional error of the end-effector can be derived as:

$$\Delta P_E^I = \begin{bmatrix} \Delta x_E^I \\ \Delta y_E^I \\ \Delta z_E^I \end{bmatrix} \approx \begin{bmatrix} 0 & -\Delta\psi & \Delta\theta \\ \Delta\psi & 0 & -\Delta\phi \\ -\Delta\theta & \Delta\phi & 0 \end{bmatrix} \cdot \begin{bmatrix} x_E^B \\ y_E^B \\ z_E^B \end{bmatrix}$$

After expanding the formula, the position error can be obtained:

$$\begin{cases} \Delta x_E^I = -\Delta\psi \cdot y_E^B + \Delta\theta \cdot z_E^B \\ \Delta y_E^I = \Delta\psi \cdot x_E^B - \Delta\phi \cdot z_E^B \\ \Delta z_E^I = -\Delta\theta \cdot x_E^B + \Delta\phi \cdot y_E^B \end{cases} \quad (4)$$

ii. Inertial force error caused by attitude angular acceleration

The angular acceleration during the attitude change of the UAV will generate an inertial force, leading to additional position errors at the end of the manipulator. According to D'Alembert's principle, the end offset caused by the inertial force is:

$$\Delta P_{E,I} = \frac{1}{k_{arm}} \cdot \begin{pmatrix} m_E \cdot a_B + J_E \cdot \begin{bmatrix} \ddot{\phi} \\ \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} \end{pmatrix}$$

Among them, k_{arm} is the equivalent stiffness of the manipulator (N/m), m_E is the mass of the end effector, a_B is the acceleration of the UAV's center of mass, and J_E is the inertia tensor of the end effector relative to the UAV's center of mass.

Considering the combined effects of attitude angle deviation and inertial force, the total position error of the end effector is:

$$\Delta P_{E,total} = \Delta P_E + \Delta P_{E,I} = [R_B^I(\Delta\phi, \Delta\theta, \Delta\psi) - I] \cdot P_E^B + \frac{1}{k_{arm}} \cdot \begin{pmatrix} m_E \cdot a_B + J_E \cdot \begin{bmatrix} \ddot{\phi} \\ \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} \end{pmatrix}$$

According to formula (4), the total position error of the end effector can be obtained:

$$\begin{cases} \Delta x_{E,total}^I = -\Delta\psi \cdot y_E^B + \Delta\theta \cdot z_E^B + \frac{1}{k_{arm}} (m_E \cdot a_{B,x} + J_{E,x} \cdot \ddot{\phi}) \\ \Delta y_{E,total}^I = \Delta\psi \cdot x_E^B - \Delta\phi \cdot z_E^B + \frac{1}{k_{arm}} (m_E \cdot a_{B,y} + J_{E,y} \cdot \ddot{\theta}) \\ \Delta z_{E,total}^I = -\Delta\theta \cdot x_E^B + \Delta\phi \cdot y_E^B + \frac{1}{k_{arm}} (m_E \cdot a_{B,z} + J_{E,z} \cdot \ddot{\psi}) \end{cases} \quad (5)$$

3) Simulation Validation of Flight Attitude Coupling

To verify the coupling effect induced by flight attitude changes and clarify the impact mechanisms of the three atti-

tudes on the stability and operational accuracy of the rotor flying manipulator system, targeted simulation experiments are conducted based on the aforementioned dynamic modeling. The simulation scenario is designed as follows: the rotor flying manipulator flies from a hovering position at 10 m to 5 m and traverses a "figure-8" trajectory. Meanwhile, the roll, pitch, and yaw angles are set to 10° respectively to simulate the flight attitude and the forces/torques acting on the manipulator base (base_link).

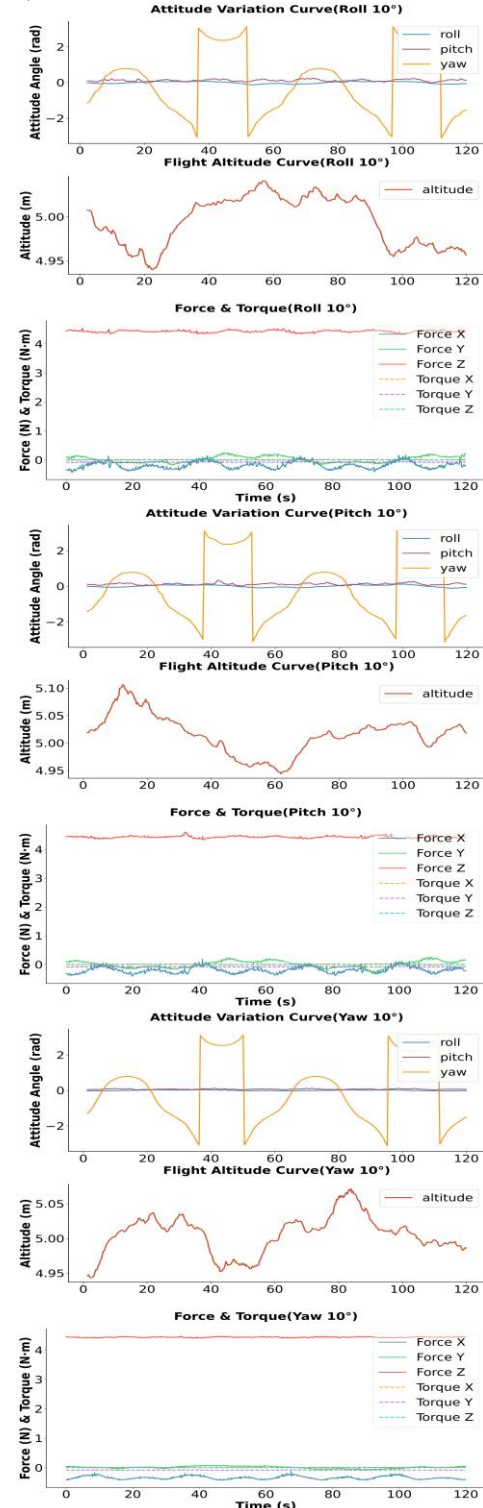


Fig. 3. Flight attitude and force response of the base.

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Table III: Fluctuation ranges of forces and torques acting on the base under different working conditions

Simulation scenario	x(N)	y(N)	z(N)	tx(N·m)	ty(N·m)	tz(N·m)
Hovering	0.02	0.01	0.01	0.00	0.00	0.00
Roll angle set to 10°	0.43	0.17	0.09	0.02	0.08	0.01
Pitch angle set to 10°	0.41	0.29	0.03	0.02	0.06	0.01
Yaw angle set to 10°	0.31	0.08	0.03	0.01	0.03	0.00

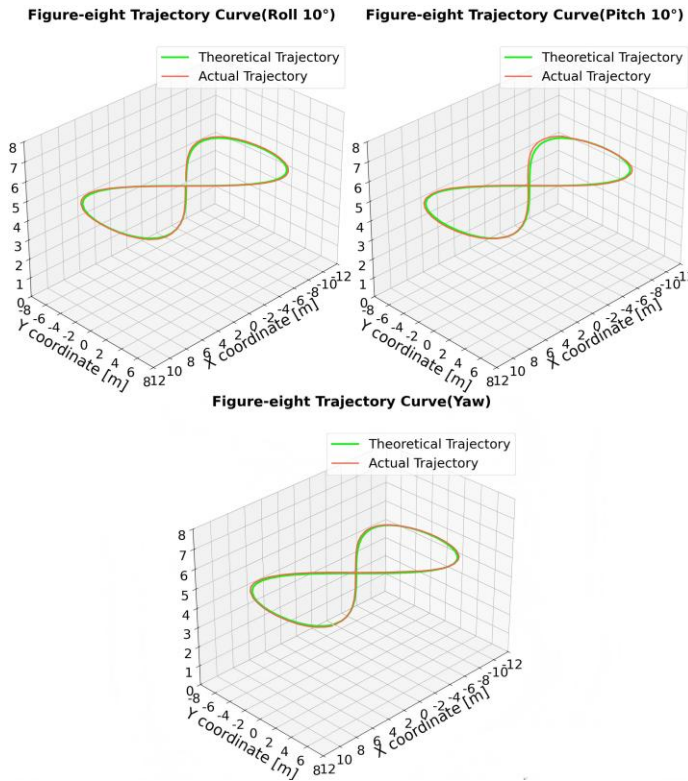


Fig. 4. Comparison of Figure-8 Trajectory Tracking.

Under different working conditions, the fluctuation ranges (peak-to-peak, P-P) of the forces and torques acting on the manipulator base (base_link) are listed in Table III.

Based on the quantitative comparative analysis of the figures and tables, the results are as follows:

(1) **Attitude Stability and Dynamic Coupling:** Under the hovering (no attitude deviation) condition, the roll and pitch angles of the system fluctuate minimally ($< \pm 0.01$ rad), while the yaw angle and altitude remain stable, verifying the intrinsic stability of the simulation platform. However, once a 10° attitude command is introduced, the fluctuations of roll and pitch angles under all three working conditions increase sharply to approximately ± 0.3 rad, and the altitude fluctuation expands to ± 0.16 m. This indicates that the system needs to continuously counteract significant coupling torques to maintain the new attitude balance. The results directly validate the theoretical analysis presented earlier: attitude changes introduce strong dynamic disturbances by restructuring the aerodynamic force distribution and inertial effects, disrupting the natural equilibrium state of the system.

(2) **Coupling Characteristics of Disturbance Forces/Torques on the Base:** As shown in the data in Table 2, compared with the hovering (no attitude deviation) condition, all attitude disturbance conditions lead to an order-of-magnitude increase in the fluctuation ranges of forces and torques on the base. Specifically, the X-axis force fluctuation increases from 0.02 N to 0.31–0.43 N (an increase of approximately 15–21.5 times), and the vertical (Z-axis) force fluctuation also increases by 3–9 times. For torques, the fluctuations of X and Y-axis torques increase to 0.01–0.08 N·m., and significant fluctuations appear in the Z-axis torque, which was nearly zero previously. Notably, pitch disturbance has the most significant impact on Y-axis force coupling (0.08N), while yaw disturbance exhibits the most prominent effect on Z-axis force coupling (4.46N). This quantitatively reveals the directional specificity of coupling effects induced by different attitude disturbances, providing precise priorities and gain adjustment basis for the subsequent design of controller decoupling compensation.

(3) **Validation of Trajectory Tracking Accuracy:** Trajectory tracking results are the ultimate manifestation of the impact of coupling effects on operational accuracy. Under the hovering (no attitude deviation) condition, the average tracking error of the "figure-8" trajectory is 0.08 m. Under the three attitude disturbance conditions, the average tracking errors increase to 0.12 m (roll), 0.15 m (pitch), and 0.12 m (yaw) respectively, with the maximum instantaneous error reaching 0.45 m, which is 2 times worse than the baseline level. This result is highly consistent with the predicted trend of the end-effector positional error model derived in Section 2.2.2 (Eqs. (4) and (5)), empirically validating that attitude coupling is transmitted through the disturbance forces/torques on the base, ultimately leading to a significant degradation in the operational accuracy of the manipulator end-effector.

The simulation experiments systematically and quantitatively validate the strong coupling effects induced by flight attitude changes from three aspects: dynamic stability, base disturbance characteristics, and operational accuracy. The data clearly indicate that even small fixed attitude commands can lead to a multiplication of system disturbance forces and a substantial decline in trajectory tracking accuracy. Furthermore, different attitude disturbances exhibit directional coupling characteristics, making it difficult for traditional linear or static compensation control strategies to effectively address such time-varying and strongly nonlinear coupling disturbances.

C. Coupling Analysis Induced by Manipulator Motion

1) Coupling Modeling of Forces and Torques Induced by Manipulator Motion

Compared with the static coupling caused by structural configuration and the directional dynamic coupling induced by flight attitude adjustment, manipulator motion is the most complex coupling source in the rotor flying manipulator system. Its joint motion directly results in the time-variation of the system's mass distribution, center of mass (CoM) position, and inertia tensor. Through the rigid connection of the base, the reaction forces and torques generated by the joint actuators are fully transmitted to the UAV platform, leading to dynamic inertial

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parameter reconstruction of the system, nonlinear superposition and transmission of inertial forces, Coriolis forces, and centrifugal forces, as well as strong time-varying interaction with the UAV attitude. Ultimately, the coupling effect exhibits characteristics of high nonlinearity, dynamic time-variation, and multi-dimensional coupling.

Due to the rigid connection between the manipulator base and the UAV fuselage, the reaction torques of the joint driving torques are directly transmitted to the UAV fuselage. Based on the Newton-Euler method and multibody system dynamics principles, the total coupling force acting on the UAV fuselage can be expressed as:

$$F_u = -\sum_{i=1}^n m_i a_i$$

$$= -\sum_{i=1}^n m_i \left[\ddot{V}_u + \omega \times r_i + \omega \times (\omega \times r_i) + J_{r,i}(\theta) \ddot{\theta} + \dot{J}_{r,i}(\theta, \dot{\theta}) \dot{\theta} + 2\omega \times (J_{r,i}(\theta) \dot{\theta}) \right] \quad (6)$$

Among them, $m_i \ddot{V}_u$ denotes the translational inertial force, $m_i (\omega \times r_i)$ denotes the Euler force, $m_i [\omega \times (\omega \times r_i)]$ denotes the centrifugal force, $m_i [J_{r,i}(\theta) \ddot{\theta}]$ denotes the joint inertial force, $m_i [\dot{J}_{r,i}(\theta, \dot{\theta}) \dot{\theta}]$ denotes the joint velocity-dependent force, and $2m_i [\omega \times (J_{r,i}(\theta) \dot{\theta})]$ denotes the Coriolis force.

According to Eq. (6), by analogy, the coupling torque can be derived as:

$$\tau_u = -\sum_{i=1}^n \{ m_i r_i \times [\ddot{V}_u + \omega \times r_i + \omega \times (\omega \times r_i) + J_{r,i}(\theta) \ddot{\theta} + \dot{J}_{r,i}(\theta, \dot{\theta}) \dot{\theta} + 2\omega \times (J_{r,i}(\theta) \dot{\theta})] \}$$

$$+ I_i \ddot{\omega} + \omega \times (I_i \omega) + J_{R,i}(\theta) \ddot{\theta} + \dot{J}_{R,i}(\theta, \dot{\theta}) \dot{\theta} \quad (7)$$

Where I_i denotes the inertia tensor of the i -th link in the body coordinate system, and $J_{R,i}(\theta)$ represents the angular velocity Jacobian matrix of the i -th link.

2) Coupled Dynamic Simulation Analysis of Manipulator Motion-Induced Coupling

To intuitively verify and analyze the coupling effects induced by the active motion of the manipulator, as revealed by theoretical modeling, dynamic simulation experiments are designed in this section. The simulation is configured such that the rotor flying manipulator descends from a hovering state at 10 m altitude to 5 m while performing a figure-8 trajectory tracking task throughout the process. The key disturbance source is set as follows: during flight, the manipulator executes randomly generated motion commands to simulate the continuous and uncertain motion of the manipulator in practical operations. This experiment aims to observe and analyze the evolution of the system's flight attitude, forces/torques acting on the base, and overall trajectory tracking performance under strong dynamic disturbances.

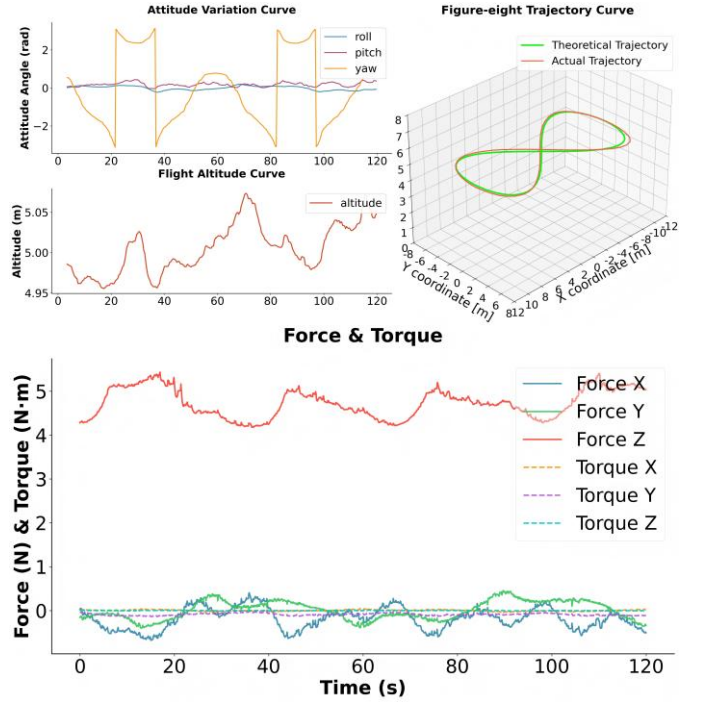


Fig. 5. Coupled working conditions under PID control.

Based on Fig. 5 (coupled working conditions) and the previous curves and data, the comparative analysis is as follows:

(1) Significant degradation of flight attitude stability: Both the flight attitude and altitude maintenance of the UAV exhibit large-amplitude, aperiodic severe oscillations, indicating that the system balance is severely disrupted.

(2) Strongly nonlinear characteristics of the base dynamic load: Compared with the hovering condition or the condition with only attitude disturbance, the forces/torques acting on the base show characteristics of sharply increased amplitude, broadened frequency band, and irregular fluctuations, indicating an increase in the intensity and complexity of coupling disturbances.

(3) Significant deterioration of trajectory tracking accuracy: The deviation of the actual figure-8 flight trajectory from the command trajectory increases, and the tracking error expands remarkably.

Simulation experiments confirm that the active motion of the manipulator is a high-intensity and complex coupling source in the rotor flying manipulator system, and the disturbances it generates are characterized by large amplitude, fast time variation, and strong nonlinearity. Therefore, developing a control strategy capable of online adapting to such complex dynamic and nonlinear characteristics is of great practical significance.

III. MULTI-AGENT-BASED CONTROL FOR ROTOR FLYING MANIPULATORS

As analyzed earlier, the strong nonlinearity and time-varying coupling existing in the rotor flying manipulator system severely limit the performance of traditional PID control methods. To address this, this paper proposes a novel Multi-Agent Actor-Critic PID (MAAC-PID) control framework. Firstly, the inherent limitations of traditional PID and existing advanced control

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methods in dealing with such complex coupling are systematically analyzed to clarify the innovative entry point of this research. Subsequently, the overall architecture of MAAC-PID, problem modeling based on Markov Decision Process (MDP), agent network design, and cooperative optimization algorithm are elaborated in detail. Finally, through high-fidelity simulation experiments, the significant advantages of the proposed strategy in terms of control accuracy, robustness, and adaptability are verified from multiple dimensions.

A. Problem Description

Although PID controllers are widely used for baseline flight control of UAVs due to their simple structure and robustness, their fixed gains are difficult to adapt to the rotor flying manipulator, an underactuated and strongly coupled complex dynamic system. Specifically, when the manipulator undergoes high-speed motion or sudden load changes, the inherent strong nonlinear coupling of the system will cause sustained oscillations, significant overshoot, and even instability of traditional PID controllers [24-28, 32-35]. In addition, the integral windup problem of PID controllers is particularly prominent when errors change rapidly, further exacerbating the degradation of control quality and becoming a major bottleneck for the system in practical dynamic operations.

To enhance the control performance of nonlinear systems, various advanced strategies have been developed. Gain-scheduling technology extends the operating range by switching PID parameter sets across predefined working modes, yet it lacks flexibility and fails to address continuously changing unforeseen conditions. Nonlinear control methods—including adaptive control, feedback linearization, sliding mode control and backstepping—boost robustness against model uncertainties or realize precise dynamic linearization via distinct mechanisms [28, 29], yielding notable outcomes in specific scenarios [30-35]. However, these approaches generally depend heavily on accurate dynamic models. For rotor flying manipulators, their model parameters vary in real time with configuration, coupled with complex unmodeled dynamics and unknown environmental disturbances. These factors impair the adaptability and reliability of the aforementioned model-dependent methods in practical applications.

Reinforcement Learning (RL) offers a novel solution to this predicament. It learns the optimal policy via continuous agent-environment interactive trial-and-error, without relying on accurate prior system models, and excels at addressing problems involving complex coupling, uncertainties, and long-sequence decision-making. RL-based adaptive PID control can automatically adjust online control parameters according to real-time system states [34-37], paving a promising way for high-performance, robust control of rotor flying manipulators. This paper innovatively integrates multi-agent reinforcement learning with adaptive PID control to design a distributed intelligent architecture, enabling online collaborative optimization of flight platform and manipulator control strategies to directly and actively compensate for the various coupling disturbances outlined in Section 2.

B. Overall Design of the MAAC-PID Framework

The core idea of the proposed MAAC-PID framework is to adopt a multi-agent cooperative paradigm of "centralized training-distributed execution". The framework consists of two independent agents: a UAV attitude control agent and a manipulator joint control agent. Both share a unified global task objective (e.g., efficiently and accurately completing grasping), but distributively output their respective control commands at the execution level, thereby ensuring global coordination while maintaining system modularity and real-time performance.

At time t , for each agent $i \in N$, under the current state $s^t = \{S_1^t, S_2^t\}$ (where $s^t \in S$), an action $a_i^t \in A_i$ is selected according to the policy $\pi_i: S \rightarrow P(A_i)$. Given the joint action $a^t = \{a_1^t, a_2^t\}$, the environment transitions from state S_t to $S_{t+1} = \{S_{1(t+1)}, S_{2(t+1)}\}$, following the state transition function P . The team obtains a reward $r_t = R(S_t, A_t)$, i.e., $R_t = \{R_{1t}, R_{2t}\}$, where R_t is the reward for agent i .

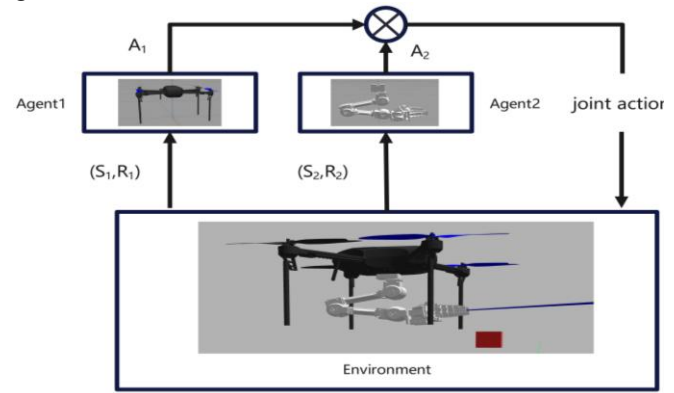


Fig. 6. Schematic diagram of the multi-agent control model.

In the centralized training phase, a centralized critic network can obtain the joint state and action information of all agents. Based on the global perspective, this network learns to evaluate the long-term value of different joint policies, thereby guiding the two agents to learn how to effectively collaborate to jointly offset coupling disturbances and optimize overall task performance. In the distributed execution phase, each agent independently generates control commands relying only on its own actor network and local observations, without the need for real-time communication, ensuring fast control response and system scalability.

This framework's innovation lies in deep integration of traditional PID control's reliability and reinforcement learning's adaptive optimization capability. As a stable, efficient low-level executor, the PID controller ensures the system's basic stability and smoothness; the upper-layer multi-agent RL module continuously learns dynamic environments and tasks to real-time adjust the low-level controller's parameters and coordination strategies, thereby actively compensating for complex coupling disturbances and adapting to diverse task requirements.

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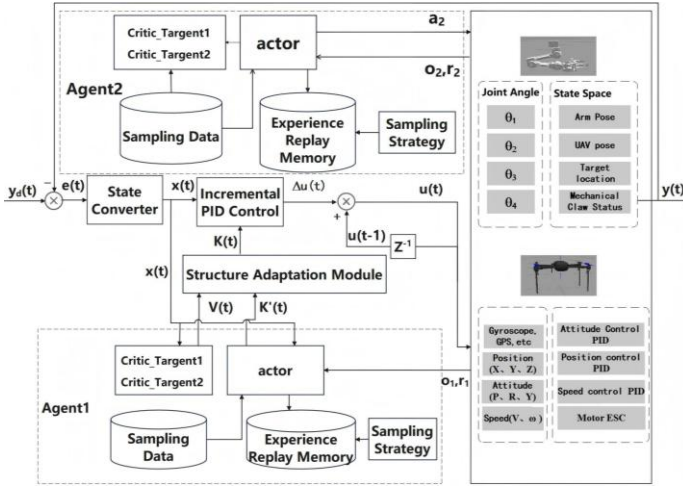


Fig. 7. Structure Diagram of MAAC-PID Adaptive Controller.

C. Detailed Design of the MAAC-PID Controller

1) Modeling of Markov Decision Process (MDP)

We formulate the cooperative control problem of the rotor flying manipulator as a partially observable Markov decision process (POMDP), which can be represented as: $\{N, S, A, P, R, \gamma, \mu\}$. Here: $N=\{1,2,\dots,n\}$ is the set of agents; S denotes the environment state space, which includes the UAV state (position, Euler angles, linear velocity, angular velocity), manipulator state (joint angles), and position of the target object; $A=\{A_1, A_2, \dots, A_n\}$ is the agent action space; P is the transition matrix, where $P(s'|s, a) : S \times A \times S \rightarrow [0,1]$ represents the probability, satisfying $\sum_{s' \in S} P(s'|s, a) = 1, \forall s \in S, a \in A$; $R=\{R_1, R_2, \dots, R_n\}$ is the reward function, where $R_i : S \times A \times S \rightarrow R$ is the reward function for agent i ; $\gamma \in [0,1)$ is the reward discount factor; and $\mu : s \rightarrow [0,1]$ is the initial state probability distribution, satisfying $\sum_{s \in S} \mu(s) = 1, \forall s \in S : \mu(s) = 0$. Assume the policy of agent i is $\pi_i : S \times A_i \rightarrow [0,1]$. Optimizing π_i is equivalent to maximizing the expected reward, i.e.: $V^{\pi_i} = E \left[\sum_{t=0}^{\infty} \gamma^t R_t(s_t, a_t, s_{t+1}) | \pi_i, \mu \right]$.

Where $\pi=\{\pi_1, \pi_2, \dots, \pi_n\}$ is the joint policy, and $R_i(s_t, a_t, s_{t+1})$ denotes the reward matrix for agent i corresponding to the joint action $a_t \in A$ at state $s_t \in S$.

i. State Space

At time t , the state of the system consists of the UAV state, manipulator joint state, target object position, and current PID parameters:

$$s_t = [U_t, \theta_t, P_t, PID_t]^T$$

where: $U_t = [p_x, p_y, p_z, \theta, \phi, v_x, v_y, v_z, w_x, w_y, w_z]^T$ represents the UAV's position, Euler angles, linear velocity, and angular velocity; $\theta_t = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5]^T$ denotes the joint angles of the manipulator; $P_t = [p_x^{obj}, p_y^{obj}, p_z^{obj}]^T$ is the coordinate of the target object in the inertial coordinate system; $PID_t = [K_{roll,p}, K_{roll,i}, K_{roll,d}, K_{pitch,p}, K_{pitch,i}, K_{pitch,d}]^T$ represents the PID parameters for the UAV's roll and pitch channels.

ii. Action Space

The actions of the rotor flying manipulator consist of the cooperative actions of the UAV agent and the manipulator agent. The action of the UAV agent corresponds to the incremental adjustments of PID parameters, denoted as: $a_1 = [\Delta K_p, \Delta K_i, \Delta K_d]^T$; The action of the manipulator agent corresponds to the adjustment increments of joint angles, denoted as: $a_2 = [\Delta \theta_1, \Delta \theta_2, \Delta \theta_3, \Delta \theta_4, \Delta \theta_5]^T$. The system's joint action vector is defined as: $a_t = [a_1^T, a_2^T]^T$.

iii. Reward Function

To guide the agents to collaboratively achieve the precise and stable grasping task, this paper designs a multi-objective hybrid reward function that integrates trajectory tracking accuracy, attitude stability, and force/torque conditions of the manipulator base. It is defined as:

$$r_t = w_1 \cdot R_t^d + w_2 \cdot R_t^s + w_3 \cdot R_t^{uav} + w_4 \cdot R_t^T + w_5 \cdot R_t^F$$

Where: R_t^d is the distance reward function, $R_t^d = -r^* |P_u - P_o|$ is the distance coefficient, and $|P_u - P_o|$ is the distance from the gripper to the target object.

R_t^s is the successful grasping reward function, defined as a piecewise function:

$$R_t^s = \begin{cases} 0 & \text{if the manipulator does not reach the target position} \\ 1 & \text{if the manipulator reaches the target position} \end{cases}$$

R_t^{uav} is the UAV attitude balance reward function:

$$R_t^{uav} = u_1 \cdot (\varphi^2 + \theta^2 + \phi^2) + u_2 \cdot (\dot{\varphi}^2 + \dot{\theta}^2 + \dot{\phi}^2)$$

Here, u_1, u_2 are weighting coefficients.

R_t^T is the force reward function for the manipulator base:

$$R_t^T = R_t^{T_x} + R_t^{T_y} + R_t^{T_z}$$

R_t^F is the torque reward function for the manipulator base:

$$R_t^F = R_t^{F_x} + R_t^{F_y} + R_t^{F_z}$$

2) Algorithm Design and Optimization of MAAC-PID

The MAAC-PID framework employs a multi-agent reinforcement learning algorithm based on the Actor-Critic architecture, while deeply integrating an adaptive PID control mechanism. Its core innovation lies in establishing a two-layer control architecture: The upper-layer multi-agent reinforcement learning module acts as a coordinator, dynamically generating optimization commands; The lower-layer parameter-tunable PID controller serves as a stable and efficient basic executor. The core of the algorithm consists of the Actor network, Critic network, target network, and the corresponding cooperative update mechanism, whose overall structure is illustrated in Fig. 9. This section elaborates on the design, cooperative mechanism, and optimization process of each module.

i. Network Structure Design

(1) Actor Network

The Actor network takes the state vector S_t of the rotor flying manipulator as input, and outputs the joint action $A = \{A_{PID}, A_{\theta}\}$, the parameter vector of the PID controller and the parameter

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vector of the manipulator's joint angles. To enhance the nonlinear fitting capability of the network, this paper designs a 4-layer deep neural network structure, which is specified as follows:

Input Layer: Receives the state vector $S_t = \{U_{it}, G_{it}, V_{it}, N_{it}, I_{it}, \theta_{it}, P_{it}, PID_{it}\} \in S$.

Hidden Layer 1: Consists of 400 neurons, employing a Batch Normalization (BN) layer and ReLU activation function: $h_1 = \text{ReLU}(BN(W_1 s_t + b_1))$.

where $W_1 \in R^{400 \times d_s}$ denotes the weight matrix, d_s is the dimension of the state vector, and $b_1 \in R^{400}$ is the bias vector.

Hidden Layer 2: Consists of 300 neurons, also using a BN layer and ReLU activation function: $h_2 = \text{ReLU}(BN(W_2 h_1 + b_2))$ where $W_2 \in R^{300 \times 400}$, $b_2 \in R^{300}$.

Hidden Layer 3: Consists of 200 neurons, with a BN layer and ReLU activation function: $h_3 = \text{ReLU}(BN(W_3 h_2 + b_3))$

where $W_3 \in R^{200 \times 300}$, $b_3 \in R^{200}$.

Output Layer: Contains d_{ai} neurons ($d_{a1}=6$, corresponding to the PID parameters of the roll and pitch channels; $d_{a2}=4$, corresponding to the relevant joint angles of the manipulator). The tanh activation function is used to constrain the output within a safe range:

$$a_i = \max_a \cdot \tanh(W_4 h_3 + b_4)$$

where $W_4 \in R^{d_{ai} \times 200}$, $b_4 \in R^{d_{ai}}$, $\max_a$ is the maximum output value vector.

(2) Critic Network

The Critic network is designed to evaluate the value of actions output by the Actor network. It takes the state and action information of all rotor flying manipulators as input and outputs the action value Q. To mitigate the overestimation bias of Q-values, a dual Critic network structure (Q_1 and Q_2) is proposed. Each Critic network adopts a 4-layer deep neural network, with details as follows:

Input Layer: Receives the joint state-action vector $[s_1, a_1, \dots, s_n, a_n]$, with a concatenated dimension of $d_{sa} = N \times (d_s + d_a)$.

Hidden Layer 1: Consists of 400 neurons, using a BN layer and ReLU activation function: $h_1^Q = \text{ReLU}(BN(W_1^Q s_a + b_1^Q))$

Where $s_a = [s_1^T, a_1^T, \dots, s_N^T, a_N^T]^T$ is the concatenated state-action vector, $W_1^Q \in R^{400 \times d_{sa}}$ and $b_1^Q \in R^{400}$.

Hidden Layer 2: Consists of 300 neurons, employing a BN layer and ReLU activation function: $h_2^Q = \text{ReLU}(BN(W_2^Q h_1^Q + b_2^Q))$.

Where $W_2^Q \in R^{300 \times 400}$ and $b_2^Q \in R^{300}$.

Hidden Layer 3: Consists of 200 neurons, with a BN layer and ReLU activation function: $h_3^Q = \text{ReLU}(BN(W_3^Q h_2^Q + b_3^Q))$.

Where $W_3^Q \in R^{200 \times 300}$ and $b_3^Q \in R^{200}$.

Output Layer: Contains 1 neuron, outputting the action value Q: $Q = W_4^Q h_3^Q + b_4^Q$

Where $W_4^Q \in R^{1 \times 200}$ and $b_4^Q \in R$.

The dual Critic networks Q_1 and Q_2 share the same structure but have independent parameters. The minimum value of the two networks is adopted to reduce Q-value overestimation.

(3) Target Networks

To enhance the stability of the algorithm, the MAAC-PID framework introduces target Actor networks and target Critic

networks. The target networks have the same structure as their corresponding online networks but are updated less frequently, serving to generate target Q-values.

Initialization: The parameters of the target networks are initialized to be identical to those of the online networks:

$$\theta^{u_i} = \theta^{u_i}, \theta^Q = \theta^Q$$

Target Q-Value Calculation: The target Q-value is computed using the target networks:

$$y_t = \sum_{i=1}^N r_{i,t} + (1 - \sum_{i=1}^N d_{i,t}) \gamma \min(Q_1', Q_2')$$

where y_t is the target Q-value at time t , $r_{i,k}$ is the reward of the i -th agent at time t , $d_{i,t}$ is the corresponding termination flag, $\gamma=0.99$ is the discount factor, and Q_1' and Q_2' are the Q-values of the dual target Critic networks, respectively.

Overestimation Bias Penalty: To further mitigate Q-value overestimation, an overestimation penalty term is introduced into the target Q-value calculation:

$$y_t = \sum_{i=1}^N r_{i,t} + (1 - \sum_{i=1}^N d_{i,t}) \gamma [\min(Q_1', Q_2') - a | Q_1' - Q_2' |]$$

Where $a=0.05$ is the overestimation penalty coefficient. By penalizing the difference between the two target Q-values, the accuracy of Q-value estimation is improved.

ii. Adaptive PID Fusion Design

The adaptive PID fusion mechanism constitutes the core innovation of the MAAC-PID controller, aiming to achieve deep coupling between multi-agent reinforcement learning (MARL) and classical PID control. By enabling online tuning of PID parameters and introducing a cooperative compensation strategy, this mechanism empowers the system to proactively adapt to the strong time-varying coupling in rotor flying manipulators. Consequently, it retains the reliability of low-level control while endowing the system with advanced cognitive and adaptive capabilities.

(1) Online Tuning and Smooth Update of UAV PID Parameters

For the online tuning of UAV PID parameters, an exponential smoothing factor is adopted to realize smooth parameter updates, thereby avoiding system oscillations caused by abrupt parameter changes:

$$K_{PID}(t+1) = K_{PID}(t) + (1 - \alpha) \Delta K_{PID}(t)$$

where $\alpha \in (0,1)$ denotes the exponential smoothing factor.

(2) Incremental PID Control Law with Coupling Feedforward Compensation

The incremental PID control law is employed to generate UAV control signals, which is defined as:

$$\Delta u_{PID}(t) = K_p(t) \cdot e(t) + K_i(t) \cdot \Delta^2 e(t)$$

Where $e(t)$ is the attitude angle error, $\Delta e(t) = e(t) - e(t-1)$ and $\Delta^2 e(t) = \Delta e(t) - \Delta e(t-1)$.

To offset the coupling torque induced by manipulator motion, a coupling feedforward compensation term is introduced:

$$\Delta u_{comp}(t) = K_{comp} \cdot J_R^T(\theta(t)) \cdot \tau_j(t)$$

Where K_{comp} is the coupling compensation coefficient, $J_R^T(\theta(t))$ denotes the angular velocity Jacobian matrix of the i -

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th manipulator link, and $\tau_j(t)$ is the joint driving torque.

The final control output integrating the incremental PID and feedforward compensation is given by:

$$u(t) = u(t-1) + \Delta u_{PID}(t) + \Delta u_{comp}(t)$$

(3) Cooperative Adjustment of Manipulator Joint Commands

To avoid conflicts between manipulator motion and UAV attitude control, the joint angle increment $\Delta\theta_j(t)$ output by the manipulator agent (Agent 2) is adjusted via a cooperative coefficient before being transmitted to the joint position controller:

$$\theta_j(t+1) = \theta_j(t) + \Delta\theta_j(t) \cdot \gamma(t)$$

where $\gamma(t) = \frac{1}{1 + \exp(-5 \cdot (R_{uav}(t) + 10))}$ is the cooperative coefficient. It can be dynamically adjusted according to the flight state and coupling intensity, and is implicitly learned by the Critic network during the centralized training phase.

IV. SIMULATION EXPERIMENTS AND ANALYSIS

A. Stability and Performance Evaluation of the MAAC-PID Algorithm

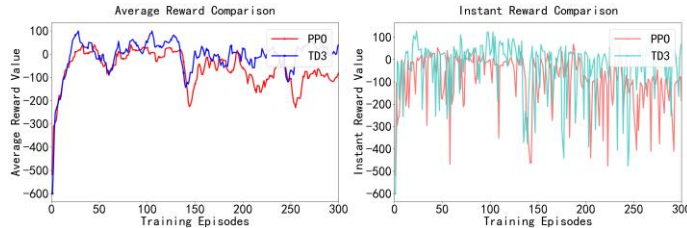


Fig. 8. Comparison diagram of training performance between MAAC-PID and PPO algorithms.

To systematically verify the control effectiveness of the MAAC-PID algorithm in dynamic coupled operation scenarios, aiming at the strong coupling disturbance between manipulator motion and unmanned aerial vehicle (UAV) flight, a simulation environment of "2 kg load + 'Figure-8' trajectory" was established. The algorithm was evaluated in terms of training convergence, steady-state control accuracy, and dynamic operation stability. Additionally, the traditional PID control strategy and the representative PPO algorithm were selected for comparative simulation analysis with the MAAC-PID algorithm. This study verifies the suppression effect of the dual-agent cooperative optimization control strategy on nonlinear coupling disturbances, thereby providing a scientific basis for the engineering application of the algorithm.

From Fig. 8, it can be observed that the training process of the MAAC-PID algorithm conforms to the convergence law of reinforcement learning: the agent gradually transitions from inefficient exploration to efficient decision-making, and the loss of the policy network (Actor-Critic) converges rapidly while maintaining stability. Additionally, the later-stage test rewards verify the generalization ability of the policy. Compared with the PPO algorithm, the MAAC-PID algorithm exhibits a significantly faster convergence speed in the middle and late training stages (after approximately 150 episodes), with the reward curve showing smaller fluctuations and the algorithm demonstrating superior stability.

Based on the analysis of the algorithm training results, after 300 training episodes, the MAAC-PID algorithm achieves a maximum trajectory error of 0.21 m and an average error of

0.09 m under a 2 kg load, while the PPO algorithm has an average error of 0.12 m. In comparison with the PPO algorithm and PID controller, the MAAC-PID algorithm maintains its error within a small range, which significantly improves the flight attitude stability.

B. Anti-Disturbance Simulation Experiment

To evaluate the anti-disturbance performance of the MAAC-PID controller, this study selects the PPO algorithm and MAAC-PID algorithm for simulation-based comparative analysis, aiming to analyze the trajectory tracking accuracy and system robustness of different control methods under various scenarios. The specific scenarios are as follows: under a load of 2 kg, when operating along the "Figure-8" trajectory and square trajectory respectively, external disturbances ranging from 0 N to 19.6 N are dynamically applied along the Z-axis direction within 60 seconds to test the anti-disturbance capability of the controllers. The results of the simulation experiments are shown in Fig. 9.

Simulation experimental results of trajectory tracking based on two typical trajectories—square and Figure-8—confirm that under scenarios with different trajectory complexities, the fluctuation amplitude of the error curve of the MAAC-PID algorithm is significantly smaller than that of the PPO algorithm. Specifically, the standard deviation of the error of the MAAC-PID algorithm in the square trajectory reduces by approximately 16.1% compared with the PPO algorithm, and it still maintains a stable error fluctuation level in the high-dynamic Figure-8 trajectory (with a standard deviation of only 0.0481). This exhibits relatively stronger system robustness and anti-disturbance capability, further verifying the adaptability of the MAAC-PID algorithm to multi-condition trajectory tracking tasks.

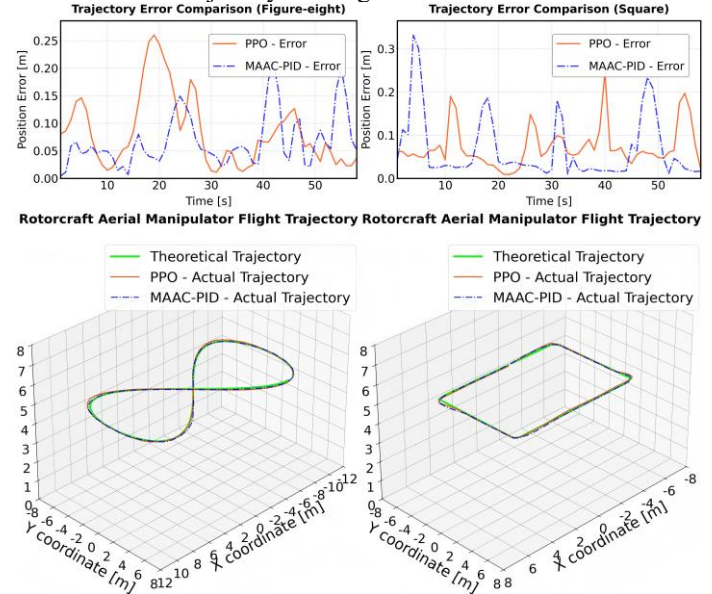


Fig. 9. Comparative results of anti-disturbance simulations.

V. CONCLUSION AND FUTURE WORK

In this paper, we have proposed an adaptive PID control strategy (MAAC-PID) based on multi-agent reinforcement learning to address the control problem of the rotor flying manipulator system. Systematic modelling has been performed on rotor flying manipulator to analyse the coupling mechanism between the rotor UAV and the manipulator. Furthermore, a design

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method for a comprehensive reward function has been also proposed, which incorporates multiple objectives. The effectiveness of the MAAC-PID controller has been verified through simulation experiments. The results show that proposed algorithms outperform when compared with existing traditional PID and PPO algorithm controller. Qualitatively, MAAC-PID controller significantly reduces the position tracking error and exhibits stronger robustness and adaptability. Quantitatively, standard deviation of the error of the MAAC-PID algorithm in the square trajectory reduces by approximately 16.1% and improvement of 0.03m average error over maximum trajectory when compared with PPO algorithm. Although, results are promising, there are certain limitations of high computational complexity, and high hardware performance which can be incorporated and implemented in future.

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